

# MARWA ALALAWI

**MASSACHUSETTS INSTITUTE OF TECHNOLOGY**  
MECHANICAL ENGINEERING

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**YEAR:** 2026  
**MONTH:** August





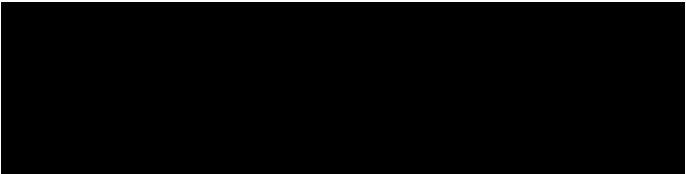
# “HELLO, I’M MARWA-”

a Ph.D. candidate at the MIT HCI Engineering Group, advised by Prof. Stefanie Mueller.

As an HCI researcher and mechanical engineer, I am interested in solving problems at the intersection of digital and physical interactions. My interests lie in interaction building, mechanical design and architecting the next generation of smart devices.

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During my PhD I worked at the University of Tokyo, where I was advised by professor Yasuaki Kakehi. I also interned as a VR developer at REACT Neuro and GREE Japan.



**YEAR:** 2026  
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**KEYWORDS:** Mechanical Engineering, Experience Design, Product Design, Electronics, Art

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### PRODUCT DESIGN: OPTICAL FASTENER SORTER

Fastener-sorting machine via optical detection and machine learning

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Hand-held Kanji to Katakana (Japanese) text convertor using optical character recognition.

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3D Printed origami-shaped molds for dumpling folding and patterning

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Thermoformed RC car with injection molded wheels

TEACHING PRODUCT DESIGN CURRICULA

Selection of product design classes I have developed & projects I have mentored

MISC MAKER PROJECTS

Miscellaneous and short-term maker projects

STAGE AND PROP DESIGN

Design for virtual and physical stage for MIT 2.s009 theme reveal; costume for Professor David Wallace; props for Bahrain-based cafe

DIGITAL EXPERIENCE DESIGN: XR & GAME DEV.

VR educational game created for GREE Japan & JAXA’s Arienai Workshop (HTC Vive); Game Development (2D , Unity)

MISC GRAPHIC DESIGN

Branding and artwork for Sakura by R&M; Artwork for MIT Japan program

SELECT 2D ANIMATION

Select 2D animations for clients using Clipstudio Paint and After Effects

COMIC ART & ILLUSTRATIONS

Compilation of comic books and short stories illustrated by me using ClipStudio Paint + Illustrator

DESIGN & FABRICATION OF INTERACTIVE OBJECTS

My research is at the intersection of digital fabrication, interactive technology, and smart materials. I focus on building smart hardware where intelligence is embedded directly into physical objects,

EMBEDDED SENSING

The 3D-printed fishing rod is printed with conductive material and controls a



RESPONSIVE DESK LAMP

Floating capacitive sensor patches detect joint rotation to dynamically adjust lighting intensity



SMART LANGUAGE TANGIBLE

Embedded mechanical state-sensing (via antenna) maps physical configuration of Chinese character block to learning UI

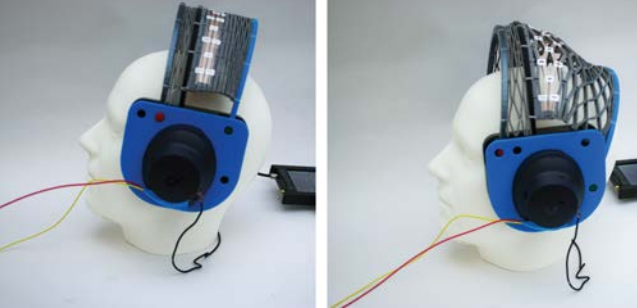


transforming everyday mechanical components into self-aware systems. Across multiple projects, I leverage multi-material 3D printing and digital fabrication to seamlessly integrate sensing, wireless communication, and energy harvesting directly into structural geometries. I also develop custom computational design tools for my systems. My work is deeply application-heavy.



RECONFIGURABLE ANTENNA CURTAIN

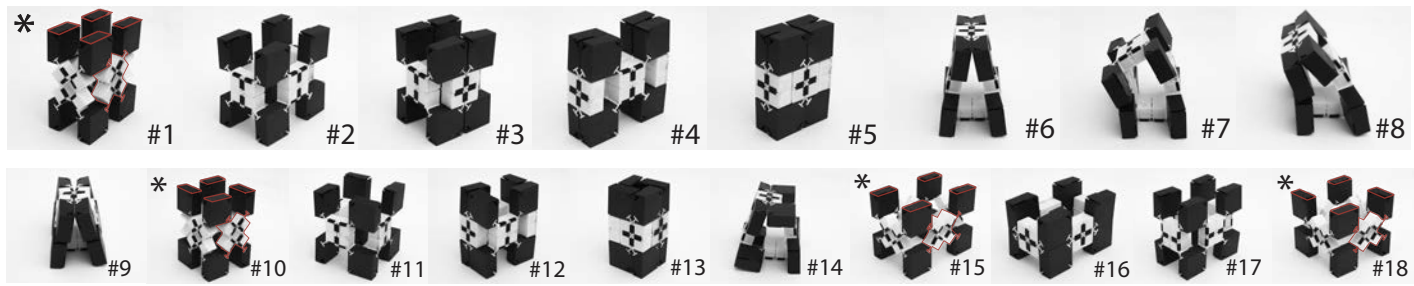
Auxetic metamaterial structures alter their shape to dynamically shift RF resonance frequencies. This curtain-embedded antenna changes the light intensity between its expanded and retracted states.



SMART HEADPHONE HELMET

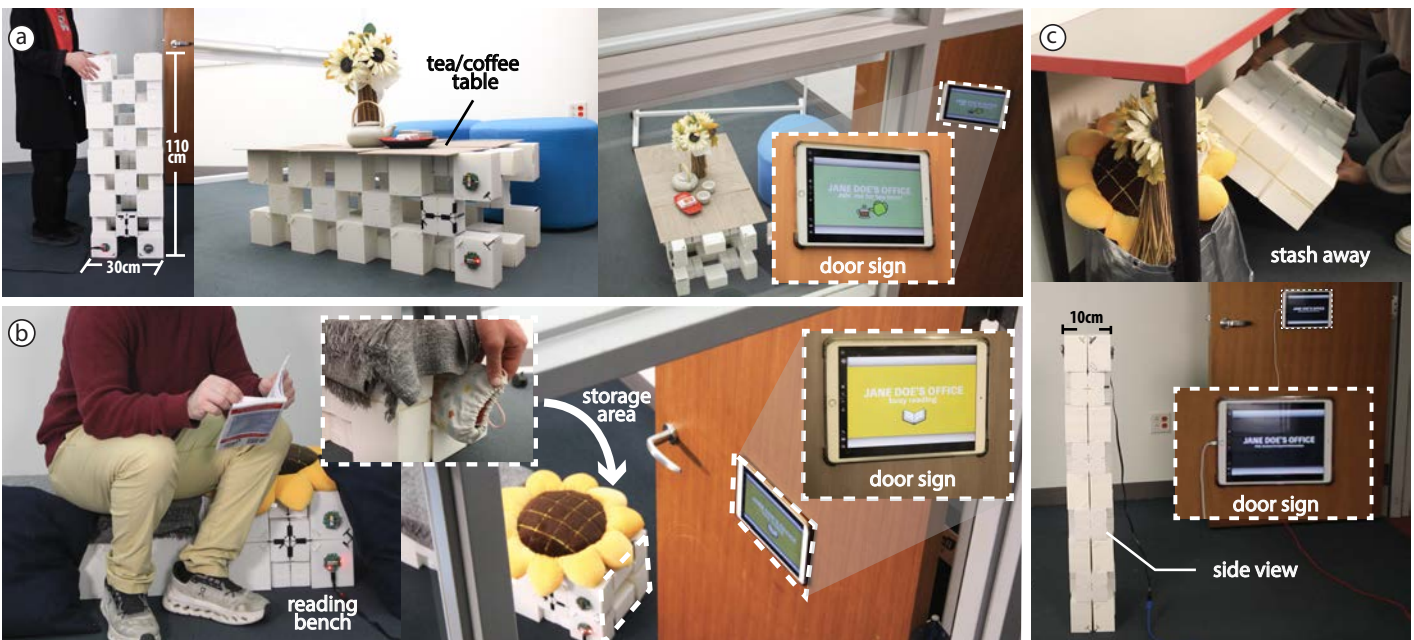
As the antenna embedded in the headphone bends and expands, the headphones change from noise cancelling to transparent mode.





## RECONFIGURABLE FURNITURE

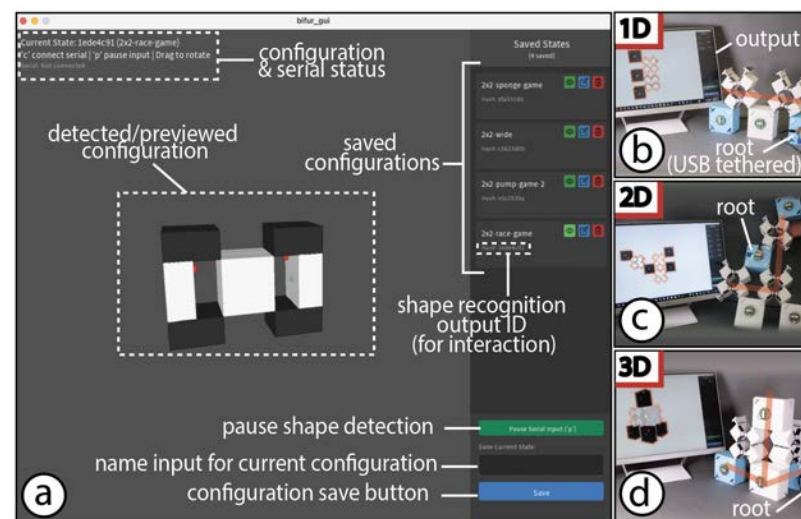
Bifur-Circuits is a 3D-printed metamaterial system with embedded conductive routing that allows shape-changing structures to self-sense their state and communicate as a collective. Bifur-circuits enable reconfigurable, human-scale furniture: (a) fully expanded as a tea table, (b) retracted with an opening as a reading bench with storage, and (c) fully collapsed for compact storage



## SHAPE CHANGING PERIPHERALS

Bifur-circuit configurations act as a shape-changing game controller, each triggering a different game.

steering wheel      sponge      pump

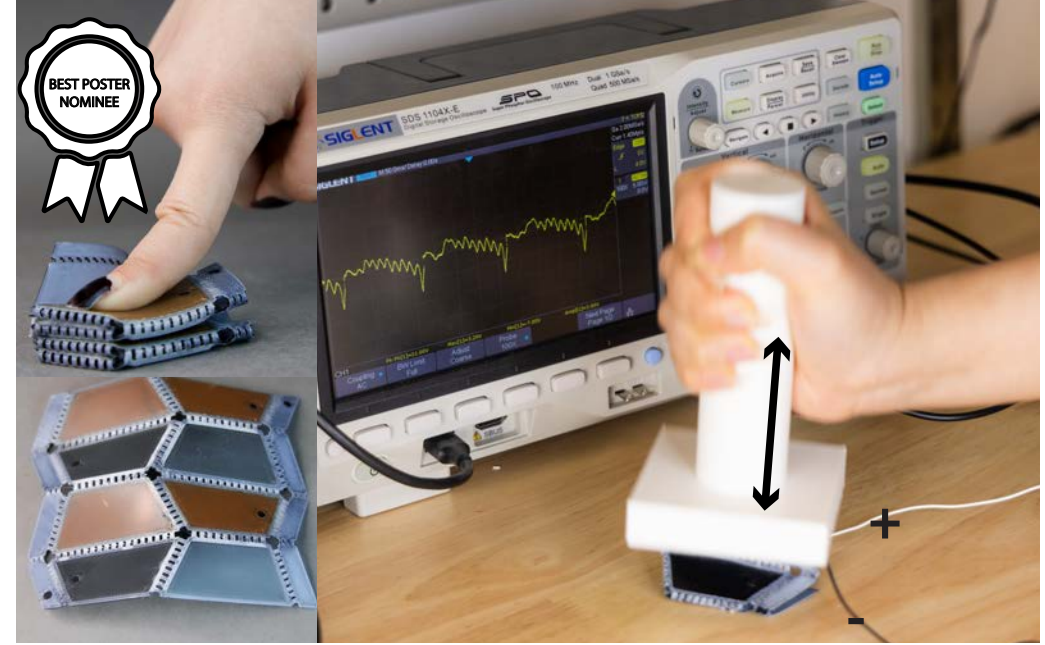


## BUILDING COMPUTATIONAL TOOLS AND UI SYSTEMS

My research delivers end-to-end pipelines that bridge software and hardware, combining custom computational UI tools for automated design and simulation. Here, Bifur-circuits, one of my projects that focuses on building electrically and mechanically modular metamaterial system has a UI that detects & saves 1D (b), 2D (c), and 3D topologies (d).

## GRADIENT THERMAL 3D PRINTING

Speed-Modulated Ironing, in collaboration with TU Delft, is a new fabrication method for programming visual and tactile properties in single-material 3D printing. We used one nozzle to 3D print and a second nozzle to reheat printed areas at varying speeds, controlling the material's temperature response, thus getting a gradient of colors.



## ORIGAMI ENERGY HARVESTING

In my project, Ori-TENG, I 3D-printed origami tessellations to function as triboelectric nanogenerators (TENGs). The geometric facets and material layering in my design ensure that when the origami pattern expands or compresses, opposing material pairs experience face-to-face contact generating static electricity as a result.



# MINION YOYOS

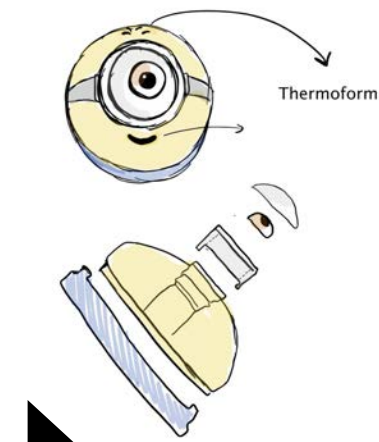
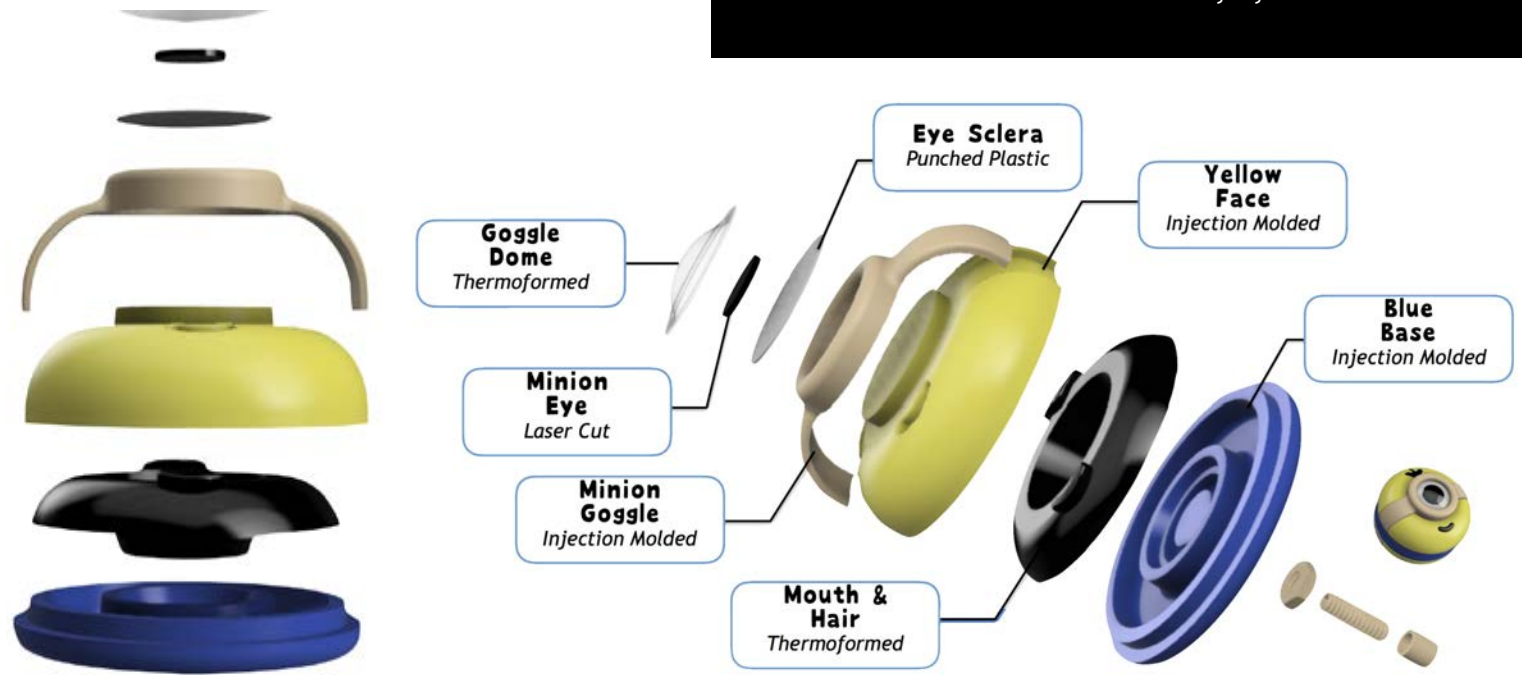
**Team Members :** Marwa AlAlawi, Benjamin Gutierrez, Lily Mueller, Jonathan Sampson

As part of MIT's 2.008 (Design & Manufacturing II), my four-person team mass-manufactured 50 functional, Minion-themed yo-yos. I directed overall product design and drove the 3D CAD, CAM, CNC mold milling, and injection molding for the yellow faceplate. Designing this component required adjusting mold dimensions for post-injection plastic shrinkage to fit both the base and goggle parts. My hands-on work also encompassed laser cutting, punching, thermoforming, and milling auxiliary molds.



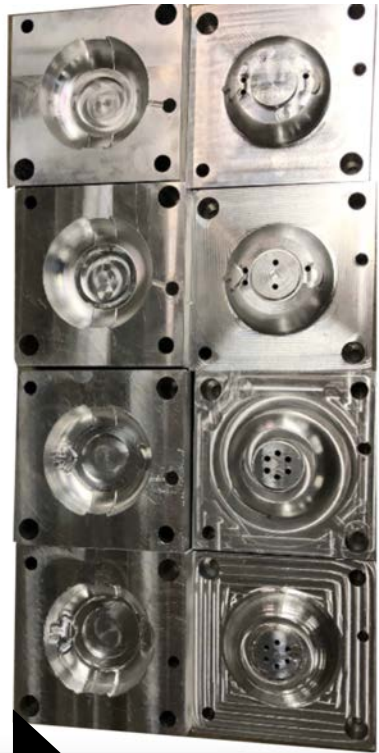
## 50 MINIONS READY

After injection molding, thermoforming, and laser cutting 50 pieces of each of the yoyo components, the yoyos were assembled by my team members and I



## SKETCH

Initial draft sketch of minion yoyo



## MOLDS

Yellow mold iterations (cavity and core). Different mold designs were created to account for plastic shrinkage as the yellow part interfaces with the blue base and goggle



## BLUE BASE

Injection molded piece of 5 degree draft angle and 1.5% plastic shrinkage



## MOUTH & HAIR

Thermoformed piece based on 3D printed die



## YELLOW FACE

Injection molded piece of 5 degree draft angle and 3% plastic shrinkage



## MINION GOGGLE

Injection molded piece with 1.5% plastic shrinkage



## SIDE VIEW

Side view of yoyo, showcasing the screw bolt



## FRONT VIEW

Front view of yoyo showcasing the sclera (thermoformed) and the eye pupil (laser cut)



# OPTICAL FASTENER SORTER

TEAM MEMBERS: Marwa Alalawi, Ben Thomson, Bruce Crawford, Darron Sandifer, Hana Khalil, Jacob Hopkins, Kevin Tang, Ne Myo Han, Seva Simone



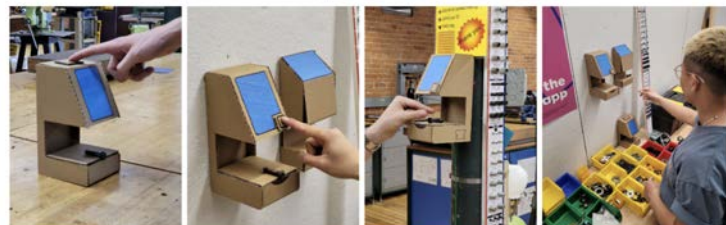
“The Nerd” is an optical fastener sorter developed for MIT Product Design & Development class

For the joint MIT + RISD Product Design & Development course, my team and I developed an automated optical fastener sorter to streamline the tedious task of organizing fasteners (nails, bolts etc) in makerspaces and workshops. As the lead software and systems engineer, I managed Raspberry Pi integration, developed the optical image-processing pipeline and sorting algorithm, and designed the UI/UX interface. During user research, makers and shop technicians highlighted key pain points, such as distinguishing between metric and imperial bolts and identifying unlabeled hardware dimensions. This prototype directly addressed our core technical question: Can we optically measure bolt diameter and thread pitch using image processing? We proved the concept successful, achieving functional dimensioning within the resolution and time constraints of the project.



## OBSERVATIONAL STUDIES

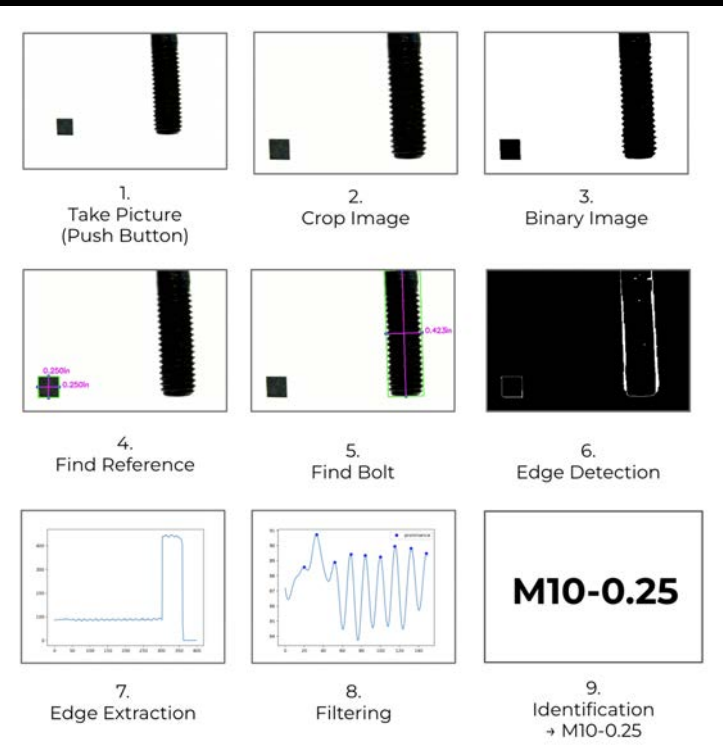
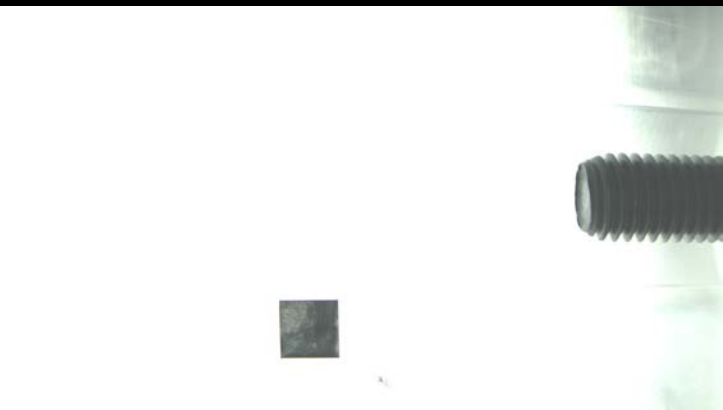
Fastener-sorting work-around solutions found in our visits to members of the maker community.



## LOW FIDELITY PROTOTYPES

Initial cardboard prototypes for our “looks-like” models.

# FASTENER RECOGNITION



My main contribution to this project was figuring out how to detect a bolt’s type via optical detection of a bolt’s diameter and pitch. To determine the bolt diameter, we used a bounding box method based on a physical reference (a vinyl cut square) of known size. For the pitch, which was relatively more complicated, we used a combination of edge detection/extraction and an FIR filtering technique (signal processing methods) to accurately determine the pitch (spacing between threads on the bolt). Finally, we had a database of pitch sizes and corresponding diameters which we used to identify the correct bolt based on a least error method from database entries. I also built the user interface using



# INTERWOVEN

**TEAM MEMBERS:**  
Marwa Alalawi, Sara Wilson, Supreetha Krishnan

Interwoven is an interactive exhibition blending art, physical computing, and engineering to explore how personal identities around sustainability evolve over time. Inspired by cycles of natural transformation, the installation emphasizes that climate action is driven by personal reflection and collective action.

### CLIMATE CRITTERS

Visitors created climate critters by stamping responses on a paper that was later hung as a "cocoon".



Each response to a sustainable question corresponds to a different design for the critter's body part.

The exhibition guides visitors through four interactive installations: (1) Creation of a Climate Critter: Visitors reflect on their environmental values to construct a custom "Climate Critter" card using laser-cut physical stamps by answering question through stamping.

### EXHIBITION SET-UP

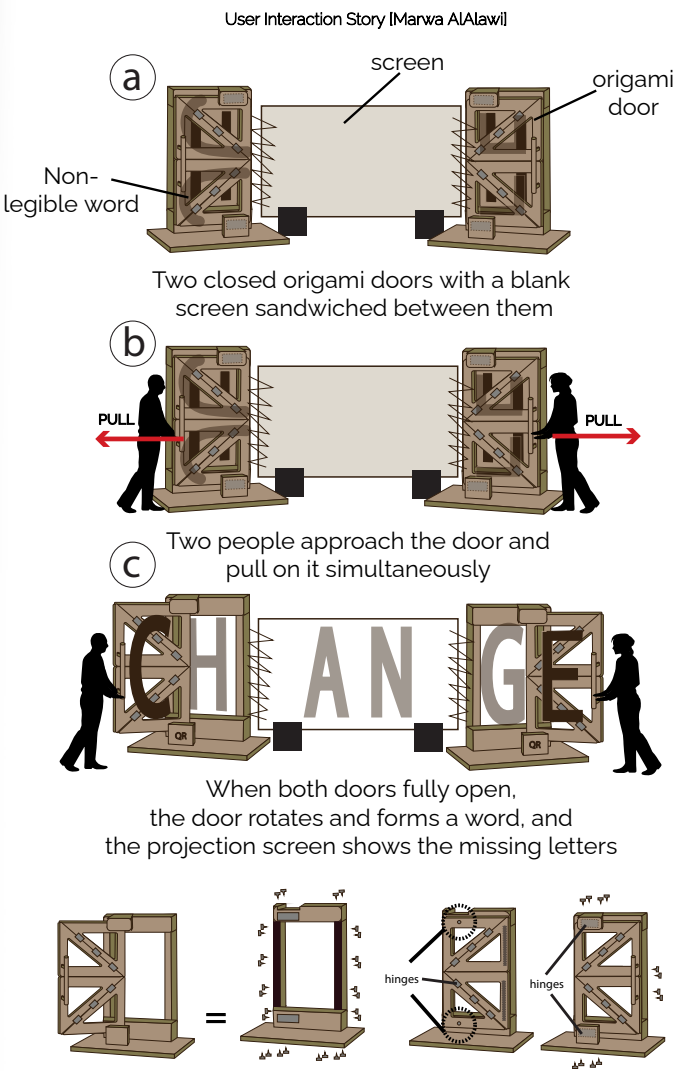
Interwoven exhibit set-up at the Weisner Student Art Gallery.



### METAMORPHOSIS (ORIGAMI DOOR)

This origami door requires two people to move it in order to have the display show the secret message.

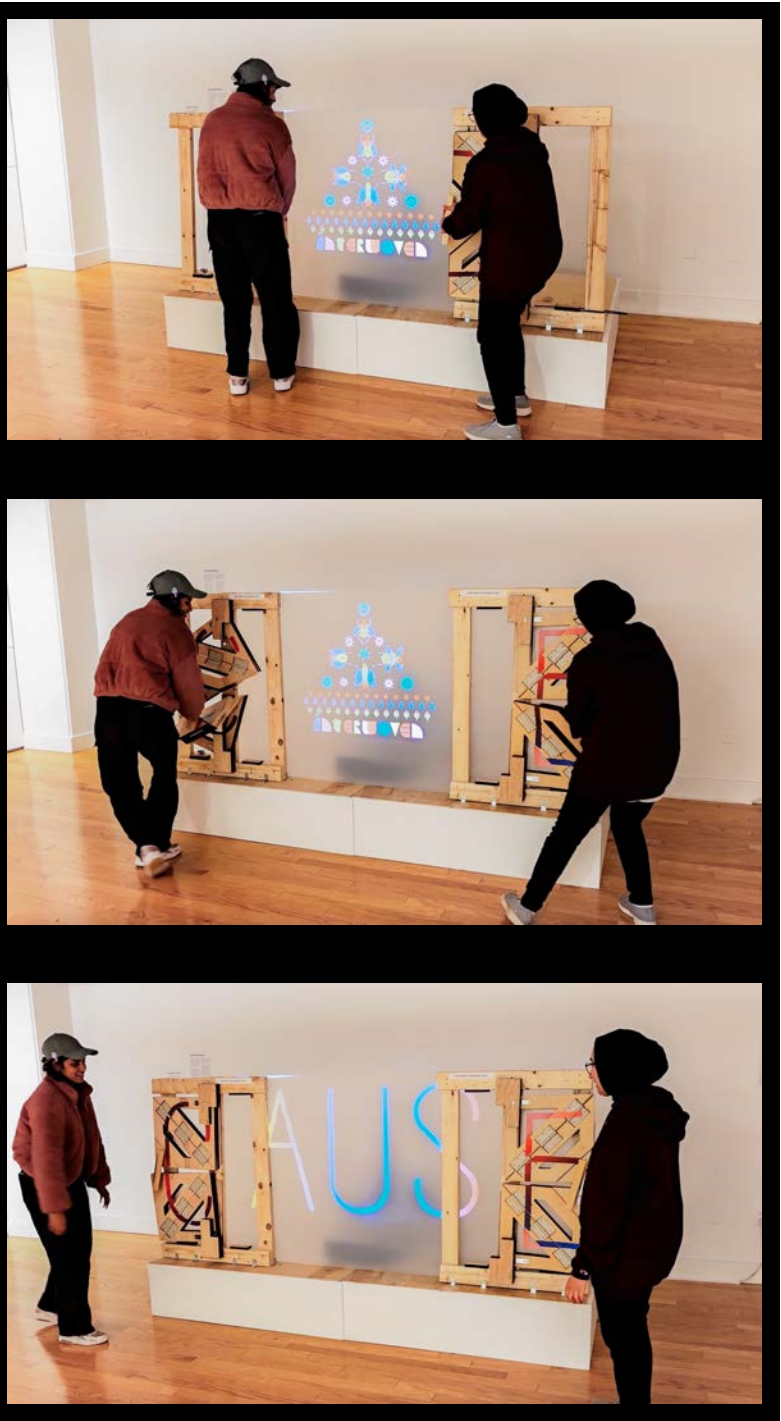
This origami door requires two people to move it in order to have the display show the secret message.



## STORYBOARD

storyboarded the mechanical principle for the metamorphosis door.

(2) The Cocoon & Weaving Station: Completed cards are hung as "cocoons," while an adjacent interactive loom allows participants to weave written reflections into a collective tapestry. (3) Critter Chimes: An interactive audio setup that translates visitor survey responses into ambient soundscapes, representing acoustic agency in climate advocacy. (4) Metamorphosis: A physical interactive mechanism where two participants must synchronously operate a shape-changing mechanism, mimicking butterfly wings emerging from a cocoon. Doing so to reveals an illuminated display reading Care, Climate, Cause.



My Contributions: Designed and fabricated the interactive, shape-changing Metamorphosis door mechanism, as well as the vector cut-files, physical stamps, and custom holders for the stamping station. I collaborated alongside Supreetha Krishnan (who designed the Climate Critters) and Sara Laura Wilson (who developed the core sustainability themes). Interwoven was featured in MIT Daily, and was supported by Arts at MIT, the MIT Council for the Arts, the MIT MindHandHeart Initiative, and the MIT Office for Experiential Learning.



# KANKAN

Handheld Real-time Translation Device for Japanese Text.

“KanKan” is an individually built device, from scratch, to aid Japanese language learners like myself through converting Japanese Kanji characters to Japanese Katakana characters. KanKan utilizes Optical Character Recognition by taking pictures of the printed Japanese kanji text, and returns on an LCD screen the same word but expressed in the phoneticized Katakana Japanese characters . For this product, I used a combination of the Tesseract OCR platform, a PCB designed by me including an ATmega, a RPI, a RPI Camera Module, and an LCD screen. The primary computer language used for coding was python, and I used it primarily for OCR manipulation as well data refinement of Japanese XML datasets. KanKan was displayed and live demoed at the MIT Media Lab, and received the praise and recognition of Media Lab Professor Neil Gershenfeld.



## FULL DOCUMENTATION

Read the full documentation of my fabrication journey of KanKan on my FabLab website: <http://fab.cba.mit.edu/classes/863.18/EECS/people/marwa/FinalProject>



## CONVERTED TEXT DISPLAY

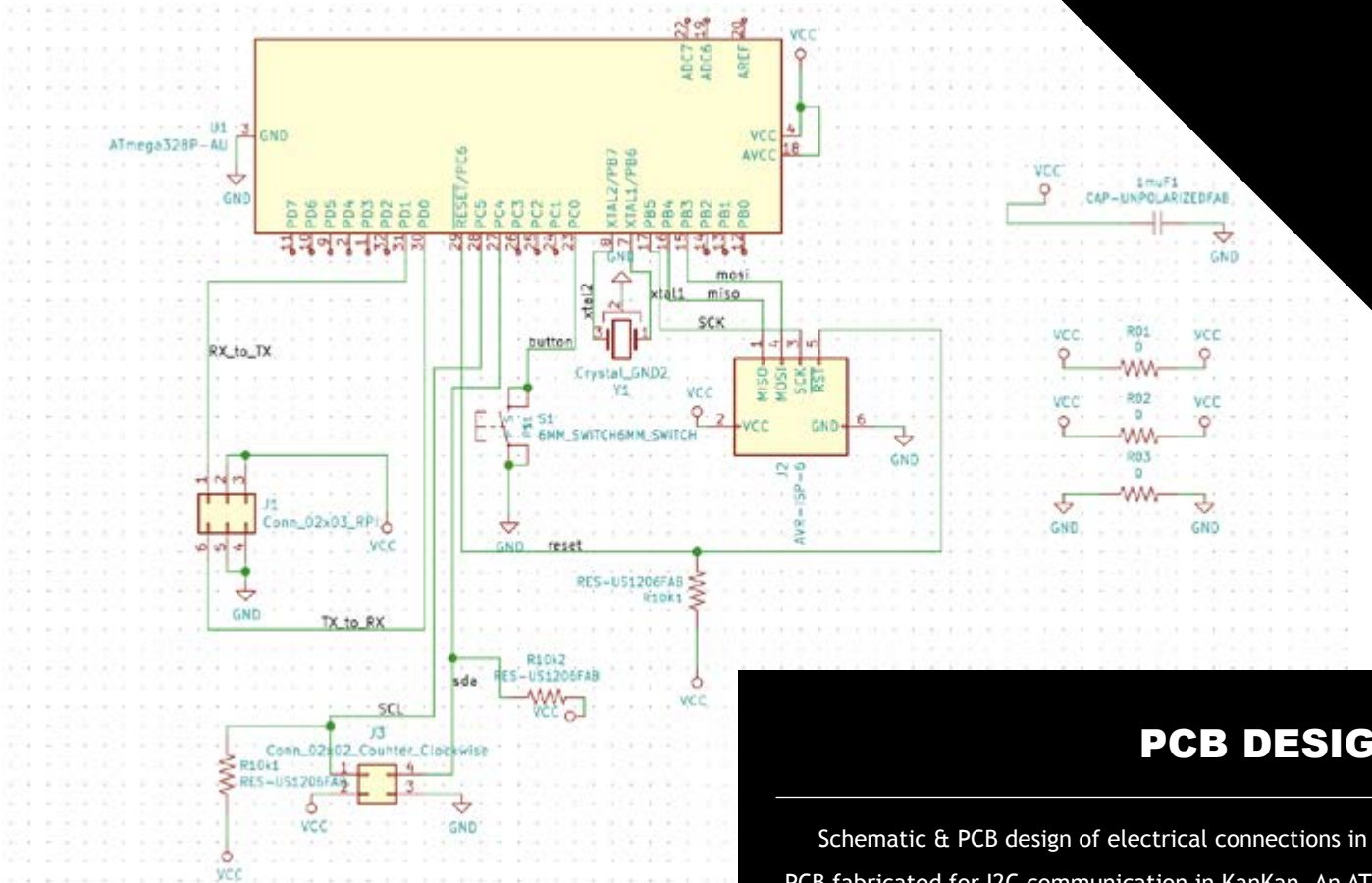
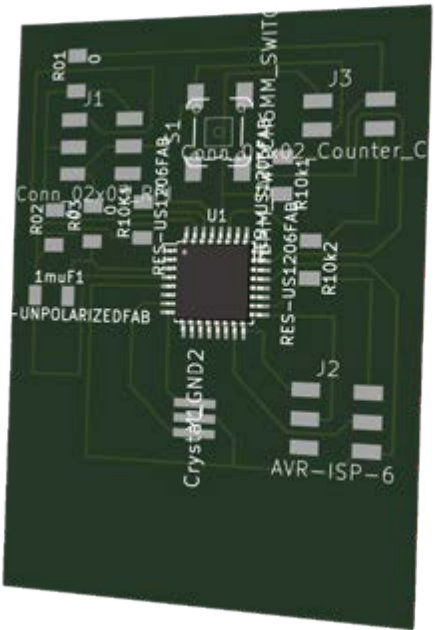
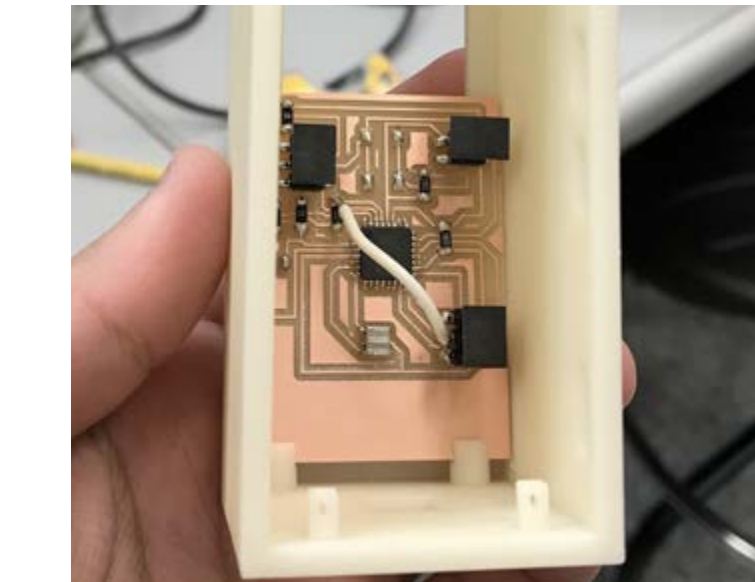
Japanese Katakana text displayed on LCD screen. RPI camera captures printed Kanji text, then OCR converts printed text to digital text that is looked up in XML dataset



## TESTING IN LAB



## DEMO AT MIT MEDIA LAB



## PCB DESIGN

Schematic & PCB design of electrical connections in the PCB fabricated for I2C communication in KanKan. An ATMEGA was used.



# MECHANICAL BEEHIVE

**Team Members :** Ben Bennington, Caroline Jordan, Devon Goetz, Fiona Mckellar, Gabe Li, Hector Castillo, Jason Ai, Kevin Zheng, Kiera Gavin, Lucy Milde, **Marwa AlAlawi**, Neel Das, Raymoung Tse, Sam Ihns, Serena Le, Val Hunter, Valerie Muldoon

2.009, also known as Product Engineering Processes, is a capstone senior project class for mechanical engineering students at MIT. In my team of 17 students, we worked on creating Aris, a mechanical beehive system that eliminates heavy box lifting from beekeeping. In traditional beekeeping that uses the langstroth hive, boxes are placed in a vertical orientation that align with natural bee activity and comfort. A hive would typically have 4 boxes on average, with the bottom two (brood boxes) housing the bees, and the upper two boxes carrying the honey. The problem arises in routine checks that beekeepers perform in order to check the two bottom brood boxes (which weight significantly less) but are in the bottom. Checking these boxes would require the lifting of the heavy upper two boxes which carry honey. This is where Aris comes in. It's a mechanical system that utilizes the concept of a four-bar linkage, and allows the boxes in the hive system to separate for easier hive access.



## 3D ANIMATION

I used the Unity Game Engine to come up with beautiful animations for our team’s product during our product pitch. I applied materials and converted our Solidworks model to a format that is animation friendly to use in Unity



## PITCHING

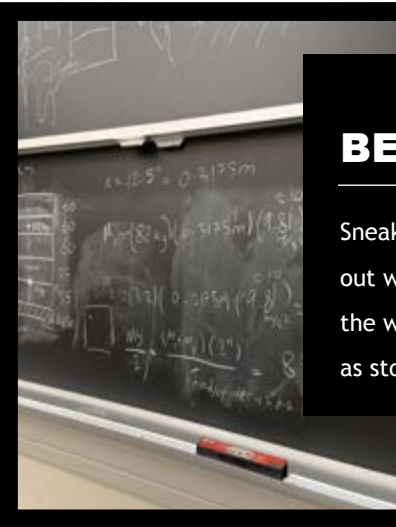
2.009 requires coming up with different product concepts. The picture here showcases my pitch on stage for one of our prototype product concepts “Port-a-pil”: a smart portable pill carrier



## ALPHA PROTOTYPE

Our alpha prototype presented at the 2.009 final presentations in front of an audience of 110 people

After coming up with different project concepts and researching them in terms of feasibility and market saturation, our team settled on creating Aris for our capstone project. I mostly worked on the mechanics and materials part of the project, where I performed several tests and calculations on failure modes, maximum loads, material selection, and determining dimensions of manufacturable parts aside from actual fabrication of the box holders. In addition, I designed the brand identity of Aris from logo to several graphics in the course, including the 3D animation shown in our final presentation, which I used the Unity Game Engine for.



## BEHIND THE SCENES

Sneak peaks from load and failure analyses that I carried out with my subteam of 2, as well as our alpha prototype in the works. Here we were experimenting with using cables as stoppers



# BAKE-ORI

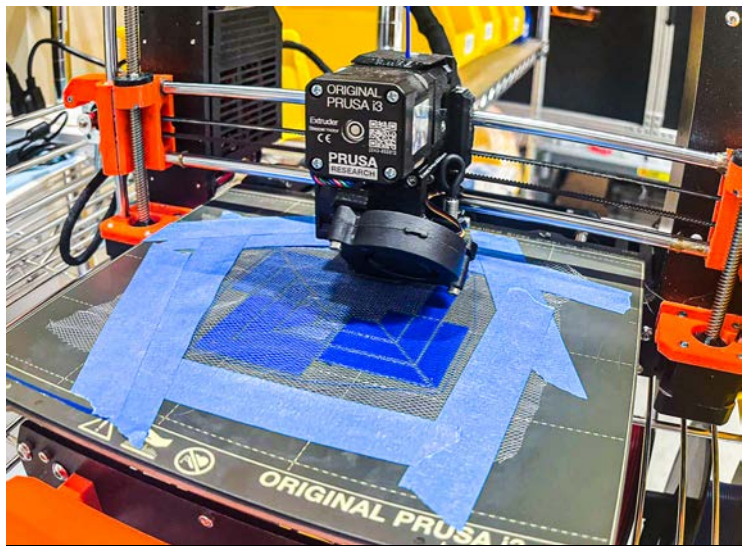
**TEAM MEMBERS:** Ashely Su, Marwa Alalawi, Cindy Yang

Bake-Ori introduces a hybrid fabrication method that overcomes the limitations of standard dumpling-folding tools by integrating origami tessellation structures with tulle cloth. My role focused on 3D modeling, optimizing print parameters for single-pass FDM fabrication,

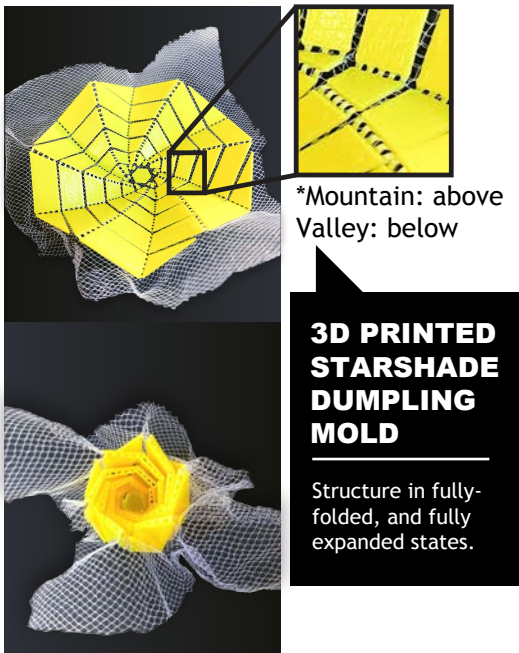
and physical prototyping to test usability across baking, frying, and steaming. Bake-Ori enables users to use molds for folding dumplings, pasta, and dough into complex geometries with simple origami tesselations 3D-printed on tulle fabric.



ORIGAMI COOKIE MADE WITH BAKE-ORI



Desktop FDM/FFF Printer Enabled



\*Mountain: above  
Valley: below

3D PRINTED  
STARSHADE  
DUMPLING  
MOLD

Structure in fully-  
folded, and fully  
expanded states.

These origami molds are more advantageous than existing tooling because they are printed flat on the print bed, and can be used on any sheet material of consistent thickness. We have also demonstrated how the same mold may be used across different cooking methods and materials,

with the material retaining its shape even after being released from its Bake-Ori mold. Bake-Ori won best demo vote at the MIT 3D printing class taught by Professor John Hart.



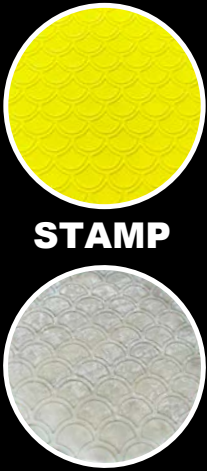
CHOCOLATE DUMPLINGS

Dumpling wrapper molded into a retracted starsjade origami pattern with Bake-Ori then air-fried to perfection.



RICE PAPER

Aside from dough, the same molds can be used on rice paper. Here we use a Miura origami mold to shape the structure. The valleys can be used to contain sauce.



STAMP

PATTERNED  
DOUGH



PATTERN PRINTING

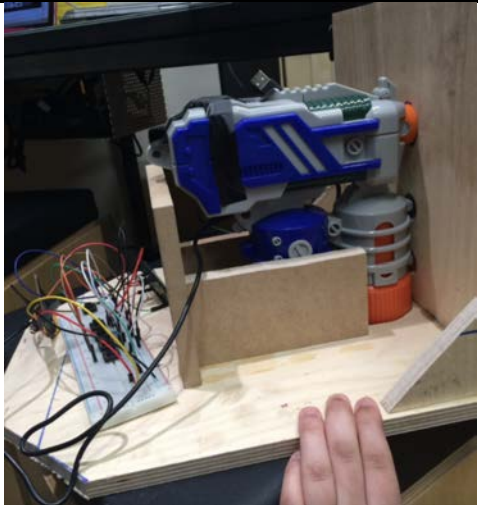
Read the full documentation of my fabrication journey of



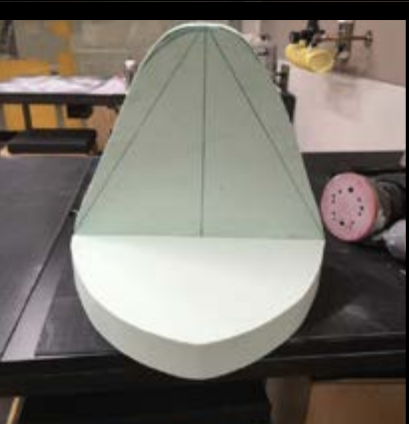
# SHARK SPLASH

Team Members : Marwa AlAlawi, Danny Gelman, Katie Henshaw, Shannon Hwang, Jake

2.00b is an introductory mechanical engineering class at MIT. In 2.00b, a team of around five students collaborate for a whole semester to build a functional toy from scratch.



Initial prototyping by hacking electronic water gun



## GREEN FOAM MOLDS FOR THERMOFORMING

Making of Shark Splash’s upper and lower mouth parts by creating green foam mold for ABS plastic thermoforming.

I worked in the team that made “Shark Splash”: a water splashing game where we randomly assign a trigger tooth to one of the shark’s many falling teeth using a teensy microcontroller. With Shark Splash, children take turns to push down the teeth from a shark’s mouth, and if by chance, the trigger tooth is pushed, the shark splashes water and the player is out of the game. The game continues until the last one standing doesn’t get any water on them by pushing down the randomized trigger tooth.

In terms of the mechanical work in the project, I was responsible for developing the teeth push and release mechanism where I made small pockets for each tooth, nested within the base, and used bendy polystyrene to allow the teeth to flexibly be pushed down, and then pulled up for resetting. I also created the green foam mold for the toy through sanding, and the final upper mouth piece of the shark. As for the electrical engineering component of the project, I was in charge of making the primary circuit connections (unsoldered iteration) and integrating the resistors, transistors and fuses. I also wrote code using Arduino for our first works-like model of the toy, where we used an electrical water gun connected to four external buttons to mimic the functionality of Shark Splash. I randomized the buttons such that any of the four buttons could potentially splash water once pressed if one of them happens to be the trigger tooth.



## SHARK SPLASH IN ACTION

Shark splash splashing water at team member Katie upon pressing of trigger tooth

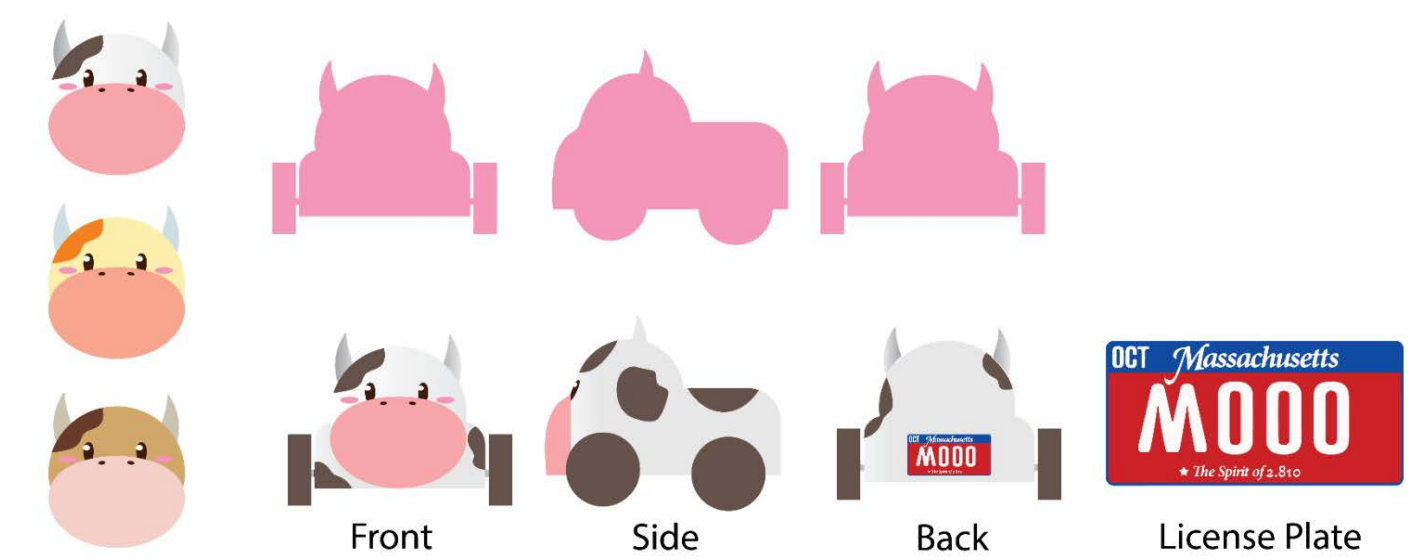


# RC CAR

For MIT's 2.810 (Manufacturing Processes), my team and I designed and manufactured six custom RC cars, earning the course "Cute Design" award for our Harvest Moon (video game) inspired cow theme. I contributed across a rapid digital-to-physical development cycle utilizing a wide array of fabrication

**TEAM MEMBERS:**  
Marwa Alalawi, Kristen Edwards, Cheng Chang,  
John Waterworth, Jonathan Bessette, Noah Bagazinski

techniques, including manual machining, thermoforming, injection molding, and SLA 3D printing. I also designed the backplate sticker for our "cow" RC car, mimicking real Massachusetts license plates.

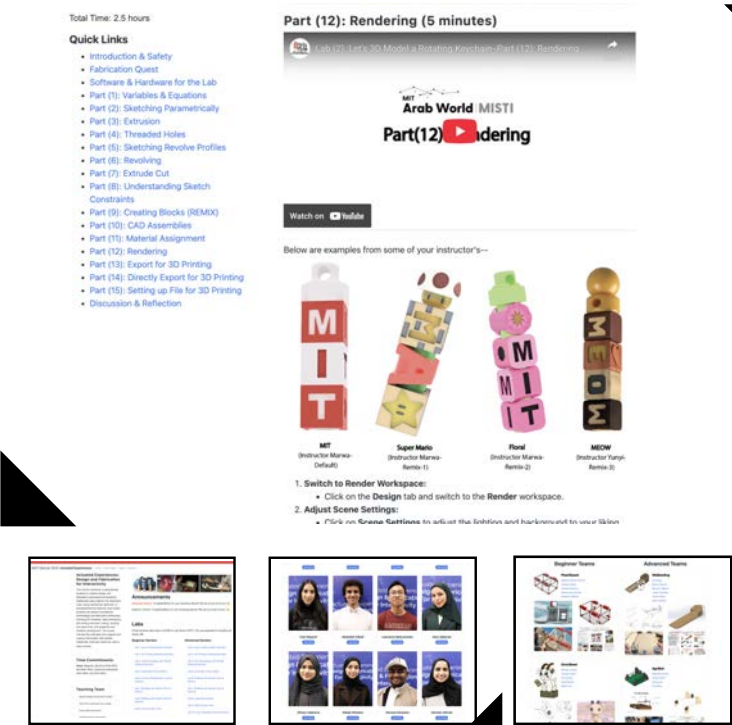


# PRODUCT DESIGN TEACHING

## ACTUATED XPERIENCES

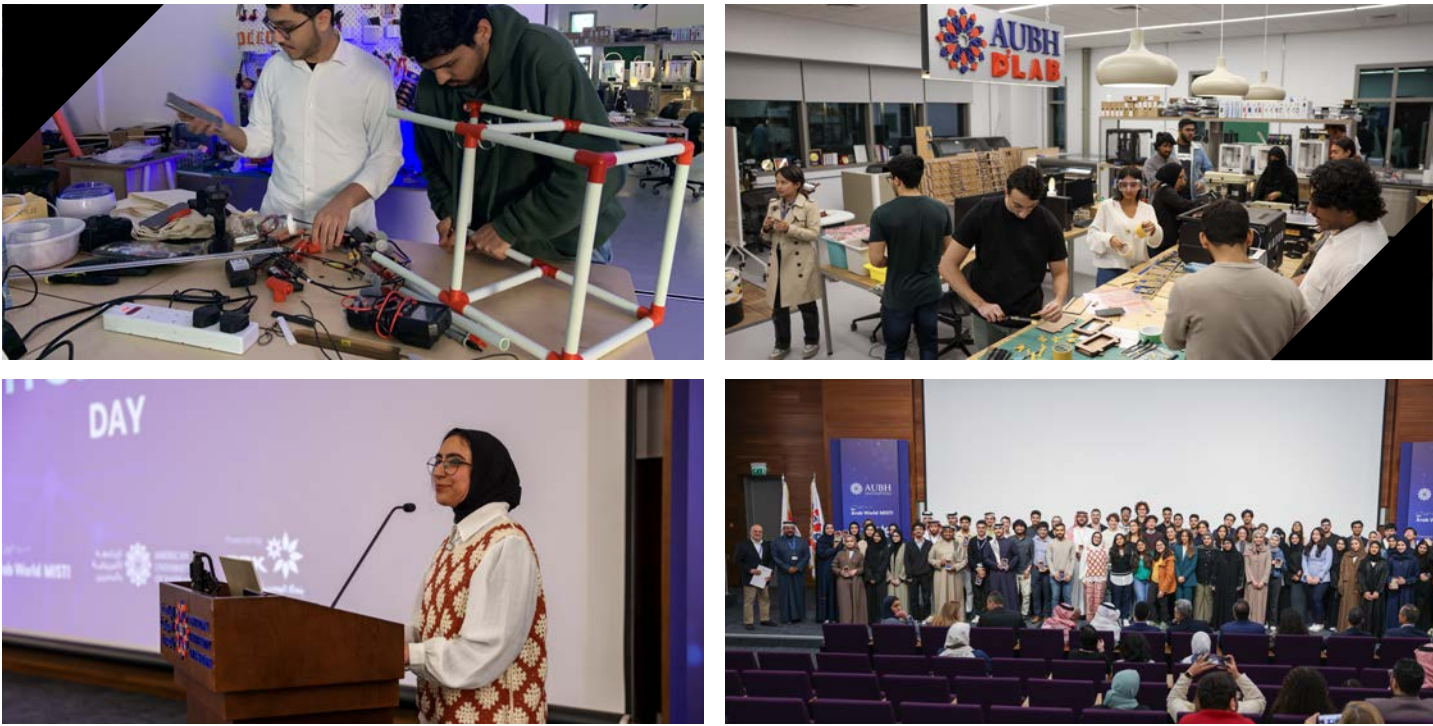
**LEAD INSTRUCTOR:** Marwa Alalawi    **CO-LEAD:** Yunyi Zhu  
**TEACHING ASSISTANTS:** Fatime Zaman, Xavier Bell, Ingie Baho

As lead course developer, I designed and led Actuated Xperiences; an intensive, hands-on interactivity course created in partnership between the MIT International Science & Technology Initiatives (MISTI) and the American University of Bahrain (AUBH). Developed for 55 beginner and advanced students, the curriculum prioritized creative application over rote learning by integrating core fabrication skills with interaction design across mechanical, electrical, graphical, and material domains. Alongside co-instructor Yunyi Zhu and a team of MIT TAs (Xavier Bell, Fatema Zaman, and Ingie Baho), we authored an open-access course portal featuring 10 structured labs, each including a custom "remixing stage" to encourage unique design variations, and guided students through five mini-projects culminating in nine final team capstone projects.



## OPEN-SOURCE CURRICULUM

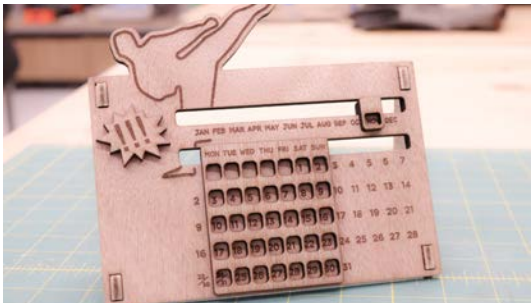
Our course website incorporated detailed tutorials (with videos and graphics) and a student portfolio section to celebrate student creations.



The program taught 55 Bahraini students from private and public universities core maker skills. The final presentation was an open-to-public live demo (with posters) similar to top-tier HCI conference demo sessions.



TUTORIAL LABS (PROJECT BASED | STUDENT REMIXES)



LAB (1)

In Lab 1: Let's Laser-Cut a Calendar!, students learn 2D CAD design in Fusion 360, basic laser cutter safety, and assembly techniques for sliding mechanical components. They build a dynamic wooden desk calendar by dimensioning components, using CAD operations (like mirroring, filleting, and linear patterning), and assembling the laser-cut pieces with joints and sliders. For the remix portion, students personalize their calendars by customizing DXF vector artwork, etchings, or modified geometric features before exporting for laser cutting.



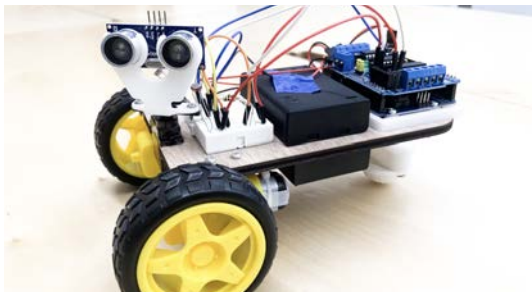
LAB (2+3)

Labs 2 and 3 guide students through the CAD design, 3D printing, and post-processing of a custom interactive rotating keychain. In Lab 2, students use Fusion 360 to parametrically model, thread, and assemble individual components before slicing them for printing; in Lab 3, they complete physical assembly and apply finishing techniques such as sanding, spray painting, vapor smoothing, and thermochromic dye application.



LAB (4)

In Lab 5: Let's Assemble a Phone Stand!, students learn essential hardware assembly techniques and power tool operations by constructing a retro TV-inspired phone stand. They practice alignment, center punching, drilling pilot holes, driving screws to join stacked laser-cut layers, and installing nuts and bolts to create functional hinged antennas.



LAB (5+6+7)

In Labs 6 through 8, students build, wire, and program an autonomous wheeled robot ("Mr. Roboto") using Arduino microcontrollers and physical computing principles. Beginning with basic breadboarding, multimeter diagnostics, and RFID interfacing in Lab 6, students progress to assembling the robot chassis and programming motor driver routines in Lab 7. In Lab 8, they transform the vehicle into a closed-loop autonomous system by integrating photoresistors and ultrasonic sensors for dynamic light-following, wall-following, and obstacle-avoidance behaviors.



LAB (8+9)

In Labs 8 and 9 students design and fabricate a custom "Guitar Hero" controller with the digital GUI through processing. For the advanced section students, they learn the principles of capacitive sensing to craft multi-input touch pads. In Lab 9, they complete the interactive system by coding a Java-based Graphical User Interface (GUI) that responds to the "touch sensing" integrated on the guitar (strings, pads of conductive materials). The GUI is based on the Guitar Hero game. The beginner section used buttons instead of capacitive touch for input.



LAB (10)

In Lab 10, students explored dynamic materials by painting a color-changing "Thermochromic Mug." Using acrylic-based thermochromic paints, they learn to layer disappearing dyes (which turn transparent above 31° C) and appearing dyes (which reveal color above 35° C) to create heat-activated patterns. To enhance their designs, students can also utilize laser-cut stencils for precise patterns or experiment with sunlight-activated photochromic dyes.

STUDENT TEAM FINAL PROJECTS

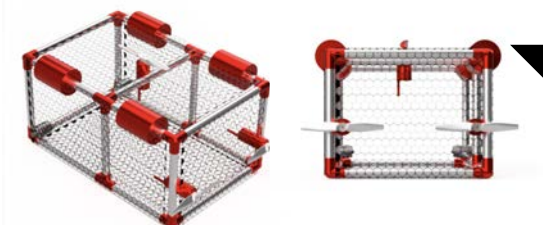
I designed the class to have final team projects with three milestones/ check-in points followed by a Final Presentation/Demo Day.



MILESTONE (1): PROPOSAL AND USER INTERACTION STORY



MILESTONE (3): FUNCTIONAL PROTOTYPE



MILESTONE (2): PARAMETRIC 3D MODEL (LOOKS LIKE)



DEMO DAY: ALPHA PROTOYPE

ORGANIZATION

Having three distinct milestones breaks the complex engineering process into a logical, phased pipeline. By demo day, all student projects were functional due to the clear organizational efforts I spearheaded with my team.

PEARL QUEST

Pearl Quest is an RC-controlled underwater drone constructed with a PVC pipe frame, propellers, and foam floats that allow it to dive and maneuver in water. It features a gate-opening mechanism to harvest Bahraini pearls from the sea floor.

MENTORS: Marwa AlAlawi & Fatema Zaman

The program brought together nine student teams (5-6 students per team) divided across Beginner and Advanced sections. Throughout the course, teams progressed through three core milestones, refining their prototypes via structured design reviews and one-on-one mentor feedback sessions.

**EMOTIBEAR**

A 3D-printed companion robot designed to support emotional learning for autistic children. By scanning tactile NFC-embedded cue cards, the robot provides multisensory feedback responses to help children explore and understand different emotions.

MENTORS: Marwa AlAlawi

**NAVCKET**

NavJacket is a wearable assistive technology to support individuals who are blind with spatial navigation and collision avoidance. Embedded with ultrasonic sensors and haptic buzzers along the left and right sides, the jacket provides directional feedback, vibrating on the corresponding side to alert the user of nearby obstacles.

MENTORS: Marwa AlAlawi

**BOOK-BUDDY**

Book Buddy is a smart digital bookmark designed to track reading progress on physical books. Featuring an embedded force sensor and an LCD screen, the device allows users to calibrate their book's total page count and sync it to an onboard ESP microcontroller, automatically updating their reading

MENTORS: Marwa AlAlawi

**MIRROR-MIRROR**

An interactive smart mirror that projects digital clothing overlays onto users, allowing them to preview outfits before a purchase. Designed for touchless navigation, the system uses ultrasonic sensors to let users swipe left or right via hand gestures to browse through clothing options.

MENTORS: Marwa AlAlawi & Xavier Bell

**AGRI-BOT**

AgriBot is an autonomous CNC watering system designed for crop care. Equipped with integrated soil sensors, the machine dynamically monitors plant moisture levels and moves along its gantry to deliver targeted irrigation where it is needed.

MENTORS: Marwa AlAlawi & Xavier Bell

**POSTURE-PACK**

Posture Pack is a smart backpack to encourage healthy posture. Embedded with back-angle sensors, the bag detects when the user begins to slouch and automatically activates a pulley and tension motor system to tighten the shoulder straps.

MENTORS: Marwa AlAlawi & Ingie Baho

**ICE-WALL**

Ice Wall is an interactive, wall-mounted art installation that utilized capacitive sensing to let users create custom pixel art. By touching designated circular nodes across the display, users can trigger instant color changes to design dynamic visual patterns.

MENTORS: Marwa AlAlawi & Ingie Baho

**WE-BOWLING**

An interactive telepresence game that translates real-world body movements into remote physical gameplay. Wearing a sensor-embedded glove, the user mimics a bowling throw using an onboard accelerometer to capture speed and trajectory, which wirelessly controls a dual-flywheel mechanism to roll a miniature bowling ball.

MENTORS: Marwa AlAlawi & Fatima Zaman







# STAGE DESIGN

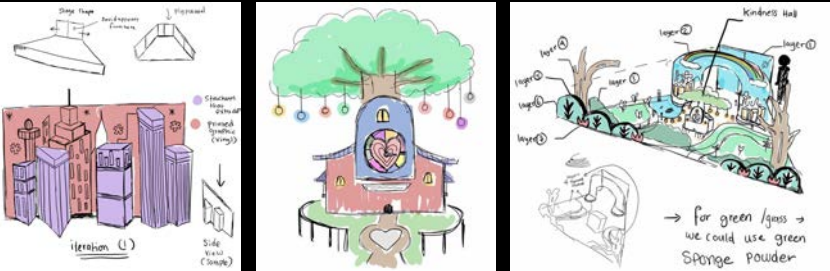
Team Members : 2.s009 Staff, ICE9 Productions

As part of my role as lead creative for stage, costume, and animation design in MIT course 2.s009, I developed the concept for the class's theme world, "Kindtopia". The stage concept I designed for the virtual theme reveal was subsequently modeled in 3D by ICE9 Productions. During initial planning, I also designed a physical stage concept for Killian Court; however, campus COVID restrictions ultimately necessitated a fully virtual event instead.



## KINDTOPIA

Original 2D Kindtopia stage design, illustrated for the theme reveal animation and used as the blueprint for the virtual event.



## STAGE DESIGN

Early iterations of the physical stage design. The concept was to cut layered plywood elements using a CNC machine



## VIRTUAL STAGE

3D rendering of the stage environment by ICE9 Productions, modeled directly from my 2D concept art.

# COSTUME DESIGN

Team Members: Georgia Van De Zande, Marwa AlAlawi

As a remote part-time teaching assistant for MIT's 2.s009, I was in charge of coming up with the design for Professor David

Wallace's Gaurdian costume for the class's theme reveal. The actual fabrication of the costume was done by Georgia Van De Zande.

## THE DESIGN



Top: sketch of costume design. Bottom: Professor David Wallace wearing the costume at the 2.s009 Theme Reveal next to past department head, professor Evelyn Wang.



## GAURDIAN

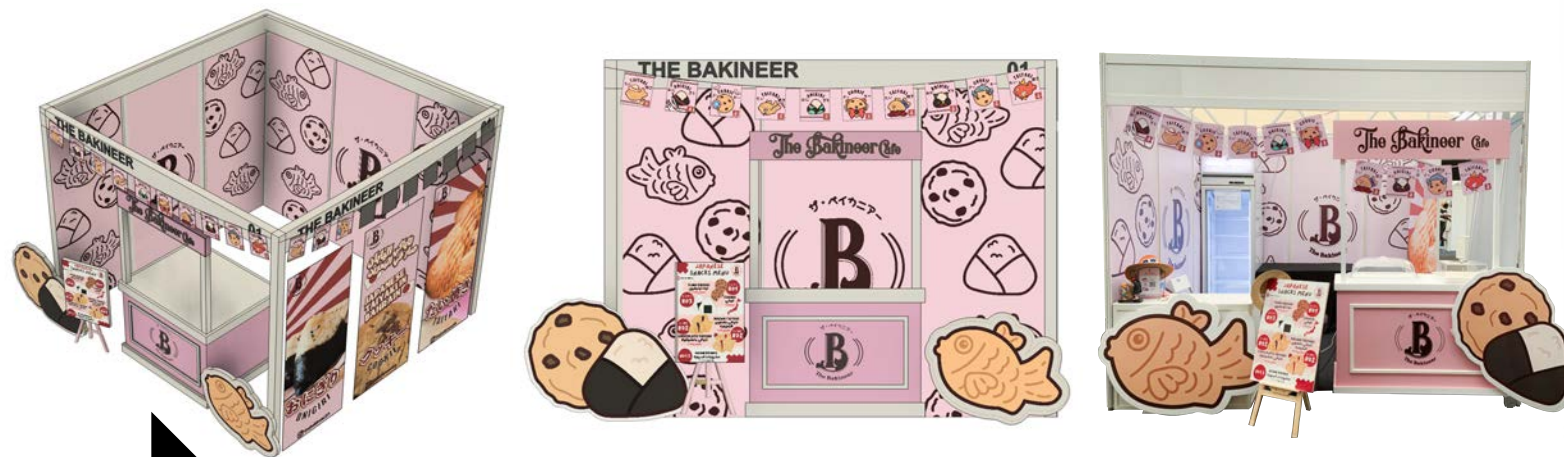
Close of up costume. Design inspired by Japanaese Gundam/Power Rangers concept.





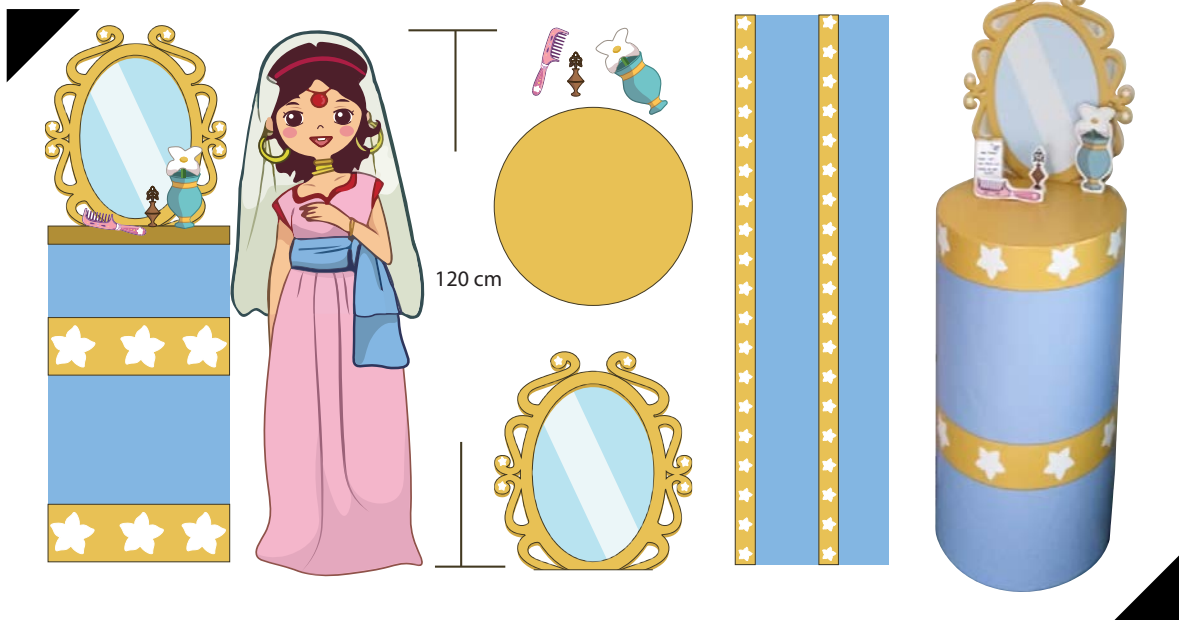
# PROP DESIGN

For ComFest 2025 (a gaming and pop culture festival in Bahrain), I designed the cart and booth space for The Bakineer Cafe. I used Fusion 360 to CAD the physical space and created accompanying custom graphics and banners that were printed and used to visually guide guests to the booth.



## 3D MODEL TO FULL DEPLOYMENT

I designed the booth in Fusion 360 with vector graphics in Adobe Illustrator; full deployment through CNC milling work.



The Bakineer Cafe in Bahrain is a themed cafe that occasionally commissions custom props for its seasonal setups. Featured here is a piece from an Arabian Nights theme, where I designed a custom princess dresser along with matching accessories and a mirror tailored to scale. The design was created using Fusion 360 and textured in Adobe Illustrator; the final image shows the fabricated prop constructed from foam core.

# DIGITAL EXPERIENCE

## DESIGN: VR



## GAME POSTER

Poster used for game promotional materials during the “Arienai” lab workshops in collaboration between GREE and JAXA.

Virtual reality educational game, "Let's Experience the Size of the Moon!"; created during a 10-week internship at GREE Japan. Developed for a joint VR attraction (Arienai Lab) between GREE and JAXA, the game educates children about lunar scale. starting with a hypothesis booth where kids guess the size of the moon compared to various Japanese landmarks. I built the game from storyboard to prototype in Unity for the HTC Vive and provided the voice acting for the character Maru-chan within the game.





# GAME DEV: INTERACTIVE VISUAL NOVEL

**TEAM MEMBERS:**  
Marwa Alalawi, Brianna Ignibosun, Muntaha Sultan

Ryou Oniisan is a visual novel game developed in Unity, based on a 2015 webcomic co-created by artists Marwa ALAlawi (Bahrain), Budoor Al Mahri (United Arab Emirates), and Muntaha Sultan (Saudi Arabia). My contributions spanned project management, overall game design and mechanics, UI/UX design, in-game script writing and storyboarding, C# coding, character and cutscene art (inking and coloring), and pixel art. The project reached a complete alpha prototype (Prototype 1) and is currently on indefinite hiatus due to the team's competing personal and professional commitments.



## LOGO DESIGN

Game logo designed by me in Adobe Illustrator (Arabic/English)



## CHARACTER SPRITES

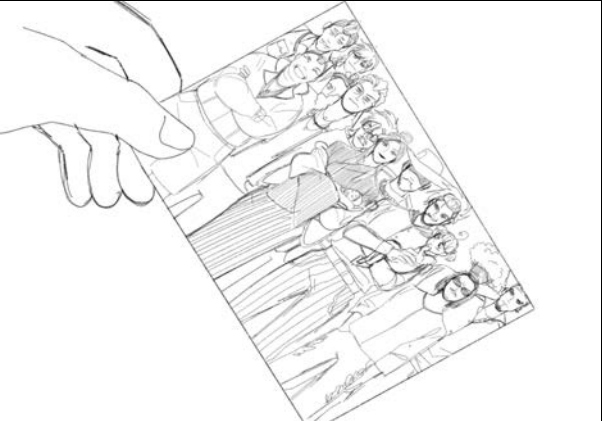
Compilation of selected character art. Muntaha Sultan was the lead on sketching character expression sprites, while I corrected the sketches, inked, and colored them; All pixel art produced by me via Clipstudio Paint.

## GAME INTERFACE



## UI Design

I was in charge of developing all in game-assets including UI elements using Adobe Illustrator.



## MINIGAMES

Our visual novel game has 6 minigames in total. I devised the concepts for all the games and created their accompanying assets/art. I coded one of the games through Unity; the rest were coded by Brianna Ignibosun.

## CUTSCENES

Compilation of cutscene graphics from game. Sketches drawn by Muntaha Sultan; Inking and coloring by me





# GRAPHIC DESIGN

## PRODUCT & PACKAGE DESIGN



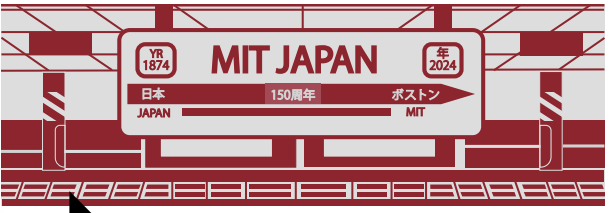
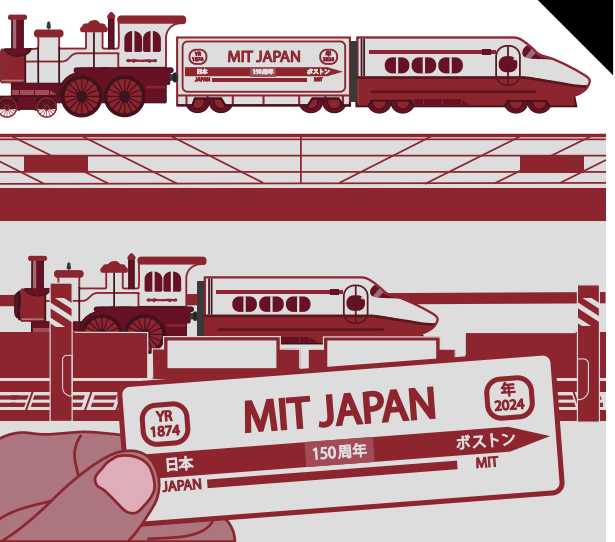
During COVID-19, I was commissioned by Sakura R&M to design, produce, and illustrate a set of double-sided baby milestone cards, complete with custom packaging. The 12-card set commemorates a baby's first 12 months. In addition to product design, I created marketing graphics for Instagram. The cards were sold locally in Bahrain at Neo Books & Coffee and The Bakineer. Prior to that, I designed the logo for the gift shop.



# GRAPHIC DESIGN

## MIT JAPAN 150th (CAMPAIGN)

I led the graphic design efforts for the MIT-Japan campaign commemorating the 150th anniversary of MIT's first Japanese graduate, Eiichiro Honma (Class of 1874, Civil Engineering). Inspired by Honma's career as a pioneering railway engineer, I designed the campaign visuals around Japanese railway maps using MIT's official cardinal red and silver-gray palette. The designs were featured on t-shirts, bookmarks, luggage tags, and stickers. My designs were worn and shared across the US, Japan, and even by Honma's direct descendants.



### EARLY AND FINAL DESIGNS

Compiltain of select early and final designs for the campaign.



### HONMA EIICHIRO GRANDSON

Members of the MIT-Japan Program and the MIT Club of Japan pictured with Eiichiro Honma's great-grandson (far left), wearing t-shirts featuring my designs.



### TIM THE BEAVER

MIT Japan Managing Director, Christine Pilcavage, with MIT mascot, Tim the Beaver, holding bookmarks featuring my designs.



# GRAPHIC DESIGN

## MIT JAPAN 40th ANNIVERSARY + NEW YEAR



### DESIGNING FOR COMMUNITY

Compilation of students (MIT, Japanese) wearing my t-shirt designs across MIT campus and Japan.

To celebrate the MIT-Japan Program's 40th anniversary, I designed a "Tim the Beaver" Daruma (a hollow, round Japanese traditional doll that symbolizes luck and perseverance). The design was printed on stickers and t-shirts, serving as that year's official MIT-Japan tshirt worn by MIT students, MIT alumni living in Japan, and Japanese students and instructors.



### TIM THE DARUMA

I designed the MIT Japan 40th anniversary logo

For New Year celebrations, I drew an MIT-themed Hagoita (a traditional Japanese wooden paddle used for New Year's good luck and ceremonial games). Featuring Tim the Beaver, the design was used on greeting cards sent to Japanese host organizations and online.



## MIT JAPAN ONIGIRI ACTION



I designed three unique graphics featuring Tim the Beaver for MIT-Japan's participation in Onigiri Action, an annual social impact campaign hosted by TABLE FOR TWO to provide school meals to children in need. The artwork was produced as badges, stickers, and enamel pins.



### TIM AND RICE BALLS

Designs turned into giveaways for event participants



My desing as a badge worn by MIT Japan & Harvard RIJS members



## MIT THEMED



Occasionally, I enjoy creating MIT-themed artwork. For an MIT student craft fair, I designed a popular sticker collection highlighting iconic campus locations and traditions. The set featured the Banana Lounge (a favorite spot offering a continuous supply of free bananas), architectural landmarks like the Stata Center, the Great Dome, and Kresge Oval, as well as the MIT ID, the Brass Rat (class ring), and MISTI (MIT's global program where students travel abroad to teach STEAM during MIT's winter term).



# ANIMATION

## 2.S.009 THEME REVEAL

Animation (1) - Animators: Marwa AlAlawi, Or Oppenheimer  
Animation (2) - Animators: Marwa AlAlawi, Or Oppenheimer |  
Background: Audrey Bosquet

Two short animations created for the MIT 2.009 theme reveal. I was responsible for storyboarding, character design, color palette selection, all key animation, and in-betweens (collaborating with Or Oppenheimer), as well as coloring and inking for half of the animated short.

Both animations can be viewed on the official MIT 2.009 YouTube Channel.



### ANIMATION (1)

The first animation was the primary theme reveal animation. I illustrated the nearly two-minute animation frame-by-frame using Clip Studio Paint, with post-editing and compositing completed in Adobe After Effects by Or.



### ANIMATION (2)

The second animation was created for the online product launch presentation, featuring students presenting on a virtual MIT Kresge Oval during COVID-19. Streamed to audiences across the US and worldwide, the event drew several thousand live viewers. I matched my visual style in this animation to Professor David Wallace's vision for a futuristic, neon-inspired world.

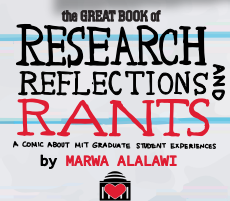
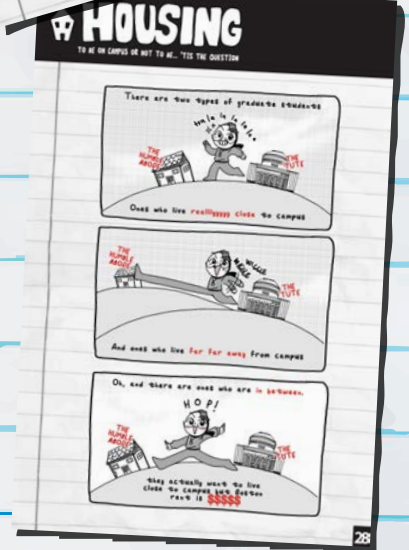
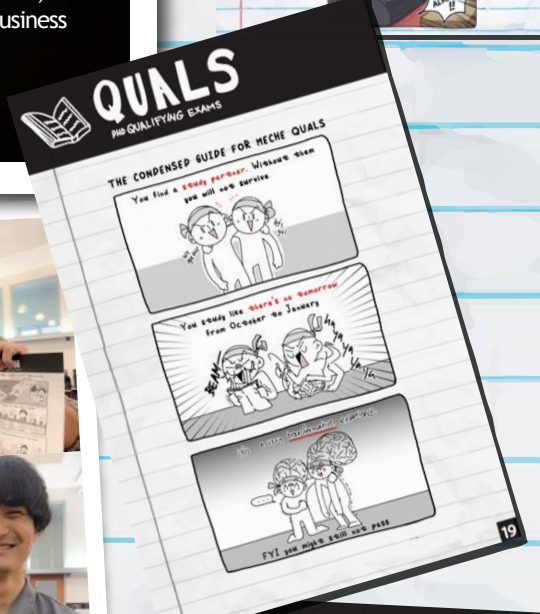
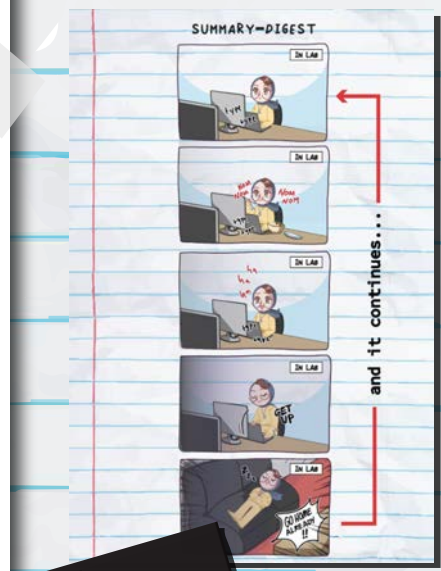
# COMIC BOOK

## RESEARCH, REFLECTIONS & RANTS



MindHandHeart

I received the MIT Chancellor Innovation Award to publish a comic book about MIT graduate student experiences. The book was published under MIT Mind Hand Heart and compiled stories from 14 students across MIT MechE, EECS, Architecture, Business & Maths.



SAY CHEESE!



# SAMPLE 2D ART

SIMPLISTIC, MANGA, CARICATURE, VECTOR



## ART RANGE

I am a self-taught (semi-professional) 2D artist and illustrator. I primarily create digital art using Clip Studio Paint and Adobe Illustrator. I am proficient in several styles, including manga art, minimalist illustration, vector art, and caricature, and have been commissioned for various projects.



## ART FOR COMMUNITY

I often volunteer to create art for my community (pro bono). For example, I drew a caricature for retiring MIT staff member Alyce Johnson and started a new tradition in my lab by drawing custom gifts for graduating PhD students.

**MARWA ALALAWI**  
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**YEAR:** 2026  
**MONTH:** August